



COMSATS University Islamabad,
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Department of Computer Science

Lab Task

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Question 1

Code :

```
import java.util.Scanner;

class QueueArray {
    int size = 5;

    int arr[]= new int[size];

    int rear = -1;

    int front = -1;

    void Enqueue(int value)
    {
        if(rear == size - 1 )
        {
            System.out.println("Array overflow");
        }
        else
        {
            if(front == -1)
            {
                front = 0;
            }
            rear++;
            arr[rear]=value;

            System.out.println("Inserted " + value);
        }
    }
}
```

```
void Dequeue()
{
    if(front == -1)
    {
        System.out.println("Array Underflow");
    }
    else{
        System.out.println("Dequeued " + arr[front]);
        front++;
        if(front > rear)
        {
            front = -1;
            rear = -1;
        }
    }
}

void Peek()
{
    if(front == -1)
    {
        System.out.println("Array is empty");
    }
    else
    {
        System.out.println("Front element: "+ arr[front]);
    }
}
```

```
    }  
}  
void Display()  
{  
    if(front == -1)  
    {  
        System.out.println("Queue is empty");  
    }  
    else  
    {  
        System.out.println("Queue elements:");  
        for(int i = front; i <= rear; i++)  
        {  
            System.out.println(arr[i]);  
        }  
    }  
}  
}  
  
public class Main{  
    public static void main( String[] args)  
    {  
        Scanner sc = new Scanner(System.in);  
        QueueArray q = new QueueArray();  
        int choice;  
        int value;
```

```
do{  
    System.out.println("1.Enqueue");  
    System.out.println("2.Dequeue ");  
    System.out.println("3.Peek");  
    System.out.println("4.Display");  
    System.out.println("5.Exit");  
    System.out.print("Enter Choice : ");  
    choice = sc.nextInt();  
    switch(choice)  
    {  
        case 1:  
            System.out.print("Enter element: ");  
            value=sc.nextInt();  
            q.Enqueue(value);  
            break;  
        case 2 :  
            q.Dequeue();  
            break;  
        case 3 :  
            q.Peek();  
            break;  
        case 4 :  
            q.Display();  
            break;  
        case 5:  
            System.out.println("Exiting");  
    }  
}
```

```
        break;

    default :

        System.out.println("Invalid Choice!");

    }

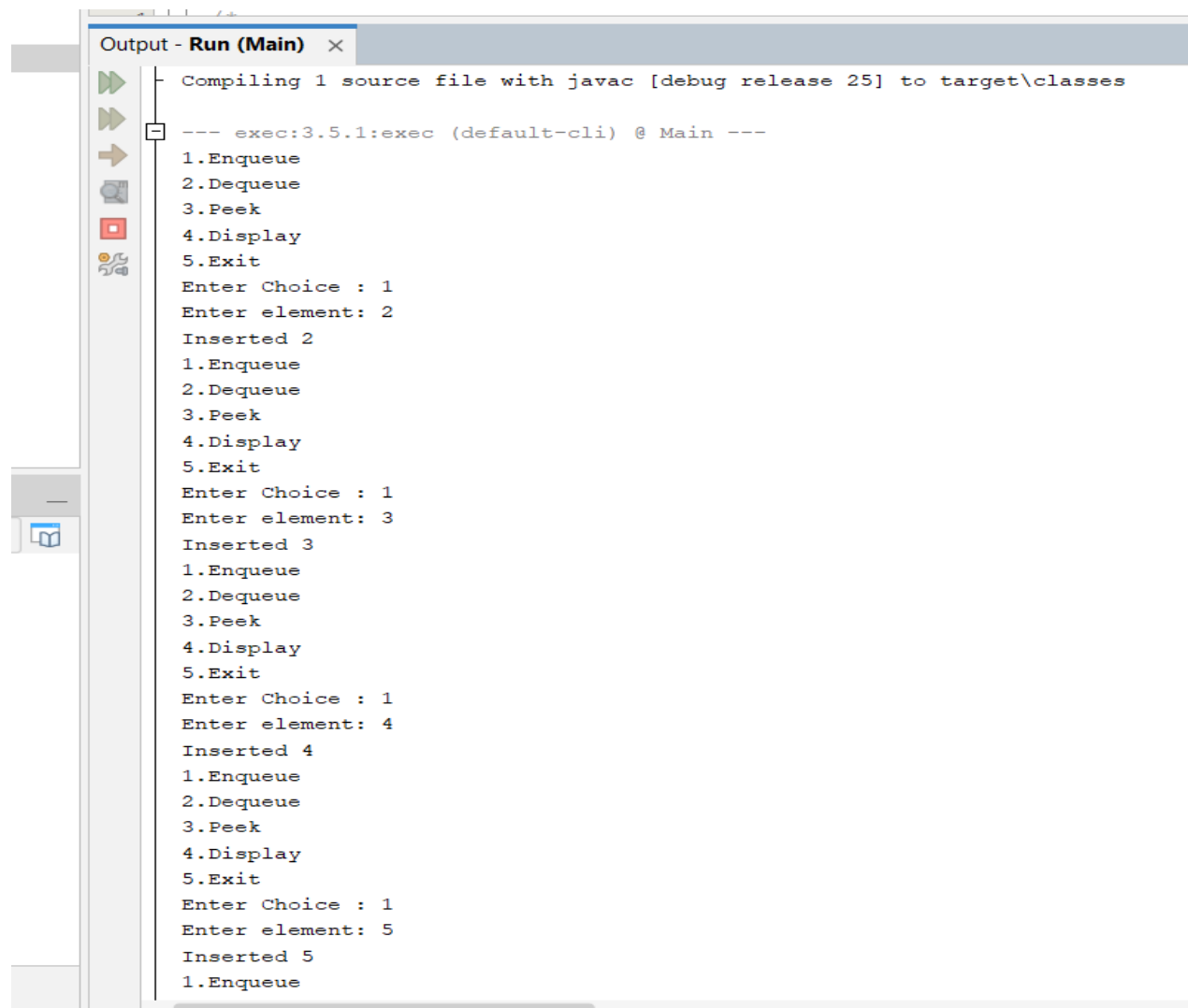
} while(choice!=5);

sc.close();

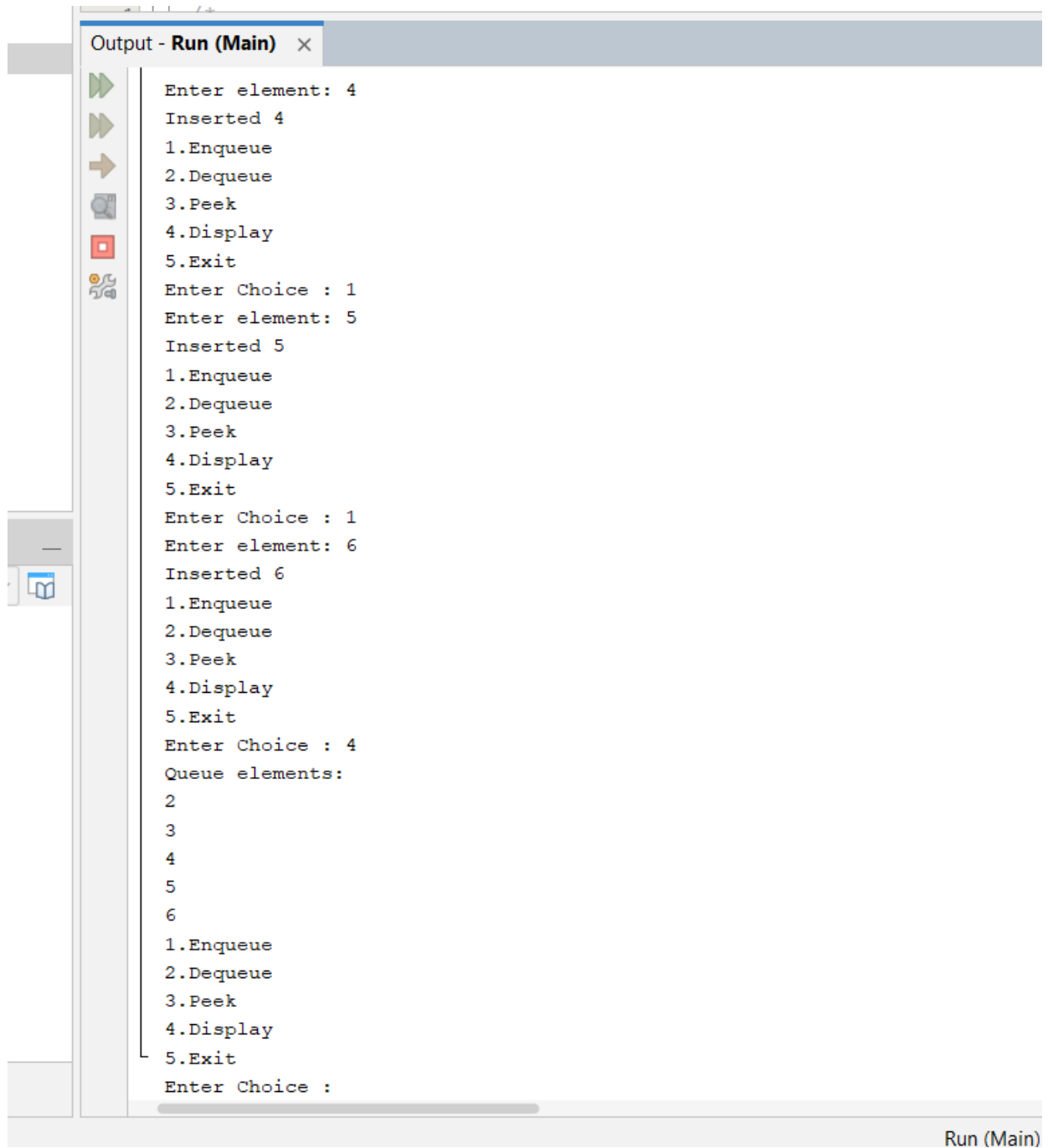
}

}
```

Output Screenshot :



```
Output - Run (Main) x
Compiling 1 source file with javac [debug release 25] to target\classes
--- exec:3.5.1:exec (default-cli) @ Main ---
1.Enqueue
2.Dequeue
3.Peek
4.Display
5.Exit
Enter Choice : 1
Enter element: 2
Inserted 2
1.Enqueue
2.Dequeue
3.Peek
4.Display
5.Exit
Enter Choice : 1
Enter element: 3
Inserted 3
1.Enqueue
2.Dequeue
3.Peek
4.Display
5.Exit
Enter Choice : 1
Enter element: 4
Inserted 4
1.Enqueue
2.Dequeue
3.Peek
4.Display
5.Exit
Enter Choice : 1
Enter element: 5
Inserted 5
1.Enqueue
```



```
Output - Run (Main) x
Enter element: 4
Inserted 4
1.Enqueue
2.Dequeue
3.Peek
4.Display
5.Exit
Enter Choice : 1
Enter element: 5
Inserted 5
1.Enqueue
2.Dequeue
3.Peek
4.Display
5.Exit
Enter Choice : 1
Enter element: 6
Inserted 6
1.Enqueue
2.Dequeue
3.Peek
4.Display
5.Exit
Enter Choice : 4
Queue elements:
2
3
4
5
6
1.Enqueue
2.Dequeue
3.Peek
4.Display
5.Exit
Enter Choice :
```

Run (Main)

Question 2

Code :

```
import java.util.Scanner;

import java.util.Stack;

public class Mainn {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Postfix Expression: ");

        String exp = sc.nextLine();

        try {

            int result = evaluatePostfix(exp);

            System.out.println("Result: " + result);

        } catch (Exception e) {

            System.out.println("Error: " + e.getMessage());

        }

        sc.close();

    }

    public static int evaluatePostfix(String exp) {

        Stack<Integer> stack = new Stack<>();

        for (int i = 0; i < exp.length(); i++) {

            char ch = exp.charAt(i);

            if (ch == ' ') continue;

            if (Character.isDigit(ch)) {

                stack.push(ch - '0');

            }

        }

    }

}
```



```
else if (ch == '+' || ch == '-' || ch == '*' || ch == '/') {
    if (stack.size() < 2) {
        throw new IllegalArgumentException("Invalid expression (not enough operands)");
    }
    int b = stack.pop();
    int a = stack.pop();
    int result = 0;
    switch (ch) {
        case '+':
            result = a + b;
            break;
        case '-':
            result = a - b;
            break;
        case '*':
            result = a * b;
            break;
        case '/':
            if (b == 0) {
                throw new ArithmeticException("Division by zero");
            }
            result = a / b;
            break;
    }
    stack.push(result);
}
```

```

else {

    throw new IllegalArgumentException("Invalid character: " + ch);

}

}

if (stack.size() != 1) {

    throw new IllegalArgumentException("Invalid postfix expression");

}

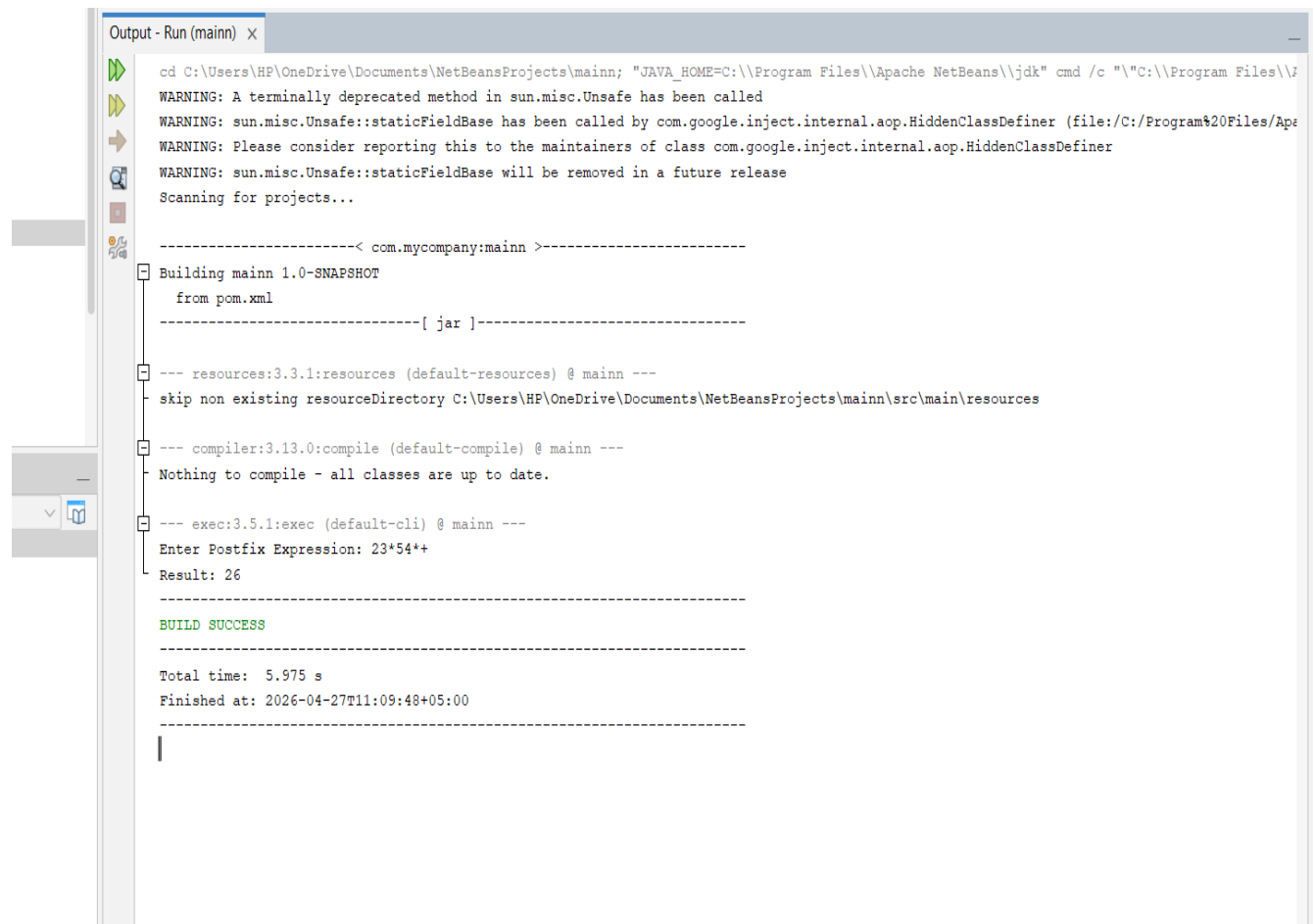
return stack.pop();

}

}

```

Output Screenshot :



```

Output - Run (mainn) x
cd C:\Users\HP\OneDrive\Documents\NetBeansProjects\mainn; "JAVA_HOME=C:\\Program Files\\Apache NetBeans\\jdk" cmd /c "%C:\\Program Files\\A
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/Program%20Files/Ape
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
Scanning for projects...

-----< com.mycompany:mainn >-----
[Building mainn 1.0-SNAPSHOT
  from pom.xml]
-----[ jar ]-----

--- resources:3.3.1:resources (default-resources) @ mainn ---
skip non existing resourceDirectory C:\Users\HP\OneDrive\Documents\NetBeansProjects\mainn\src\main\resources

--- compiler:3.13.0:compile (default-compile) @ mainn ---
Nothing to compile - all classes are up to date.

--- exec:3.5.1:exec (default-cli) @ mainn ---
Enter Postfix Expression: 23*54*+
Result: 26

BUILD SUCCESS

Total time: 5.975 s
Finished at: 2026-04-27T11:09:48+05:00

```